Module-1 2m

1. Define data preprocessing. Mention any two steps involved in the data preprocessing process.
2. What is NumPy? State any two advantages of using NumPy arrays over Python lists.
3. What is Pandas? Differentiate between a Series and a DataFrame.
4. Define imputation. Mention any two methods used to handle missing values in a dataset.
5. Differentiate between feature scaling and normalization.
6. What is a Z-score? Write the formula used to calculate it.
7. Define the Interquartile Range (IQR). Write the formulas used to identify the lower and upper bounds for outlier detection.

5m

1. Explain the major steps involved in data preprocessing and discuss their importance in preparing data for analysis.
2. Explain NumPy array creation, indexing, slicing and basic arithmetic operations with suitable Python code examples.
3. Describe Pandas Series and DataFrames. Explain how to create a DataFrame, inspect its contents and handle missing values using Pandas functions.
4. Explain different imputation techniques, including mean, median and mode imputation. Discuss when each technique is appropriate.
5. Explain the Z-score and IQR methods for outlier detection. Compare their procedures and identify situations in which each method is useful.

Module-2 2m

1. Define feature engineering and state any two of its techniques.
2. Define the correlation coefficient and mention its range of values.
3. What is the purpose of the Chi-square (\(\chi^2\)) test in feature selection?
4. Define eigenvalues and eigenvectors of a matrix.
5. What is an elementary transformation of a matrix? Give one example.
6. Define Principal Component Analysis (PCA).
7. State any two advantages of PCA in dimensionality reduction.

5m

1. Explain feature engineering and describe its major techniques with suitable examples.
2. Explain the correlation coefficient and calculate Pearson’s correlation coefficient for a given dataset.
3. Explain the Chi-square test and its role in selecting categorical features.
4. Explain eigenvalues and eigenvectors and describe their significance in data analysis.
5. Explain elementary row and column transformations of matrices with suitable examples.

Module-3

1. What is Matplotlib? Mention any two types of plots supported by it.
2. What is a line plot? State its use.
3. Define a bar chart and a histogram.
4. What is a scatter plot? When is it used?
5. What is a pie chart? Mention its purpose.
6. What is a box plot? How does it help identify outliers?
7. What is Seaborn? Mention any two advantages of using Seaborn.

5m

1. Explain how to create a line plot and a bar chart using Matplotlib with suitable Python code.
2. Explain how to create a histogram and a scatter plot using Matplotlib. Mention their applications.
3. Explain how to create a pie chart and a box plot using Matplotlib with suitable examples.
4. Describe the basic features of Seaborn and explain how to create a scatter plot and a bar plot using it.
5. Explain different types of data visualization plots and discuss their uses with suitable examples.